EXECUTIVE SUMMARY

U.S. Federal Medicaid Drug Utilization & Cost Intelligence

A 10-Year State-Level Analysis (2016–2024)

Document Title	Executive Summary — U.S. Federal Medicaid Drug Utilization & Cost Intelligence
Project Reference	Medicaid Drug Spend Intelligence & Cost Optimization
Prepared By	Bhumi Nagar — Business Analyst
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Primary Stakeholder	Medicaid Managed Care Organization (MCO) & Payer Analytics Team
Tools	PostgreSQL | Power BI | DAX | SQL | Lucidchart
Data Source	CMS Medicaid State Drug Utilization Data (Medicaid.gov)
Data Coverage	17K+ drugs | 50+ states | 2016 - 2024

1. EXECUTIVE OVERVIEW

The Medicaid Drug Spend Intelligence & Cost Optimization project was developed to analyse large-scale Medicaid drug utilization and reimbursement data across the United States — transforming fragmented multi-year healthcare data into a centralized business intelligence solution. The project was designed to simulate a real-world healthcare analytics implementation lifecycle, combining structured business analysis, SQL-based data engineering, KPI governance, and executive dashboard reporting.

The final solution addresses a critical visibility gap faced by Medicaid managed care organizations: despite managing billions in pharmaceutical spend annually, most payer analytics teams lack the structured infrastructure to identify what is driving those costs — and where to intervene.

2. BUSINESS PROBLEM STATEMENT

Medicaid managed care organizations and payer analytics teams routinely face the following challenges:

- Limited visibility into rising pharmaceutical spending trends over time
- Difficulty identifying the specific drugs and states driving the majority of costs
- No standardized KPI framework for monitoring spending, utilization, or cost efficiency
- Fragmented, disconnected data across multiple yearly files — no unified analytical structure
- Lack of reimbursement model comparison (MCOU vs FFSU) to inform payer strategy
- Reactive rather than proactive cost management — decisions made without adequate analytical support

This project addressed these challenges by building a structured analytics solution that converts raw CMS Medicaid data into actionable business intelligence.

3. SOLUTION APPROACH & METHODOLOGY

The project followed a structured 3-phase business intelligence lifecycle:

Phase 1	Phase 2	Phase 3
Business Analysis	**Data Engineering (SQL)**	**Dashboard Development**
· Project Charter	· Raw data ingestion (COPY)	· Power BI semantic modelling

· BRD (FR-001 to FR-021) · KPI Definition Document · As-Is / To-Be Analysis · Process Flow Diagrams	· Cleaning & standardization · Suppression handling logic · Feature engineering (4 KPIs) · 11 analytical KPI views	· DAX measures & KPI cards · 6-page executive dashboard · Pareto & trend analysis · State benchmarking visuals

4. DATA SOURCE & SCOPE

Source	Coverage	Records	Models
CMS Medicaid State Drug Utilization Data (Medicaid.gov)	2016–2024 · 50+ states + District Columbia	17,000+ drug-level records across 10 yearly files	MCOU (Managed Care) and FFSU (Fee-for-Service)

Key fields used: Drug Name · State · Utilization Type · Total Amount Reimbursed · Medicaid Amount Reimbursed · Number of Prescriptions · Units Reimbursed · Suppression Flag

5. KPI FRAMEWORK

A structured KPI governance framework was developed across 6 analytical categories — 15 KPIs total:

Category	KPIs
Financial	Total Spend · Medicaid Spend · Non-Medicaid Spend
Utilization	Total Prescriptions · Total Units Reimbursed
Efficiency	Avg Cost per Prescription · Avg Cost per Unit · Avg Units per Prescription
Dependency	Medicaid Dependency Ratio (Medicaid Spend / Total Spend)
Concentration	Spend Share · Prescription Share · Cumulative Spend Share (Pareto)
Data Quality	Suppression Rate · Suppressed Rows · Non-Suppressed Rows

6. KEY ANALYTICAL FINDINGS

The following findings were identified from analysis of 10 years of federal Medicaid drug utilization data:

01. Cost Concentration Risk
~20% of drugs drive ~80% of total Medicaid pharmaceutical spends across 50+ states — a classic Pareto 80/20 pattern indicating high formulary concentration risk.

02. Consistent Spending Growth
Medicaid drug spending showed a strong, consistent upward trend from 2016 to 2024, reaching $1.5T total spend. 2025 data excluded due to partial-year reporting.

03. Managed Care Dominance
~90%+ of total Medicaid pharmaceutical expenditure flows through Managed Care Organizations (MCOU) — structural dependency on managed care reimbursement confirmed.

04. Geographic Concentration
California, New York, and Texas account for a disproportionate share of total spend — geographic outliers requiring a targeted cost containment strategy.

05. Price-Driven Cost Growth
Average cost per prescription stands at $307 — rising faster than prescription volume, confirming cost increases are primarily price-driven rather than utilization-driven.

06. Data Quality Considerations
Suppression rates vary by year, state, and utilization type. FFSU records have higher suppression rates than MCOU — documented and managed through SQL transformation logic.

7. BUSINESS RECOMMENDATIONS

Based on analytical findings, the following cost optimization recommendations are proposed for the MCO payer analytics team:

1. **Target high-cost drug tier for formulary review:** The top 20% of drugs driving 80% of spend should be prioritized for formulary management, therapeutic substitution review, and rebate negotiation — particularly high-cost specialty biologics identified in the Pareto analysis.
2. **Investigate price-driven cost growth:** With average cost per prescription at $307 and rising faster than volume, MCOs should assess pharmaceutical pricing agreements and benchmark against industry rebate standards to identify overpayment risk.
3. **Geographic cost containment strategy:** CA, NY, and TX represent disproportionate spend concentration. State-specific formulary and utilization management programs should be evaluated for these high-impact markets.
4. **Evaluate MCOU vs FFSU cost efficiency:** With 90%+ of spend flowing through managed care, MCOs should assess cost-per-prescription performance between MCOU and FFSU models to identify where each reimbursement structure delivers better value.
5. **Deploy continuous KPI monitoring:** The Power BI dashboard should be deployed as an ongoing monitoring tool — tracking spending trends, cost per prescription, and Pareto concentration quarterly to enable proactive cost management decisions.

8. BUSINESS VALUE DELIVERED

Analytical Capabilities

- Centralized analytical visibility into 10 years of drug spend
- Standardized KPI framework — 15 metrics across 6 categories
- Reusable SQL transformation pipeline (Raw → Staging → Analytics)
- Executive-ready Power BI dashboard for non-technical stakeholders
- State-level benchmarking across 50+ jurisdictions

Business Impact

- Identified ~20% of drugs driving ~80% of total spends
- Confirmed ~90%+ managed care dominance in Medicaid spending
- Flagged CA, NY, TX as high-priority geographic intervention targets
- Determined cost increases are price-driven, not volume-driven
- Delivered 5+ actionable cost optimization recommendations

9. PROJECT DELIVERABLES

#	Deliverable	Description	Status
1	**Project Charter**	Project scope, objectives, stakeholders, success metrics, risks & assumptions	☑ Complete
2	**Business Requirements Doc (BRD)**	FR-001 to FR-021, NFR-001 to NFR-005, KPI definitions, acceptance criteria, current state analysis	☑ Complete
3	**KPI Definition Document**	15 KPIs — business context, formulas, interpretation guidance, design principles, future enhancements	☑ Complete
4	**SQL Data Pipeline**	PostgreSQL ETL: raw → staging → enriched → analytics — 11 KPI views, suppression logic, feature engineering	☑ Complete
5	**Process Flow Diagrams (4)**	Lucidchart: As-Is/To-Be, BA Workflow, End-to-End Data Workflow, Dashboard Architecture Pipeline	☑ Complete
6	**Power BI Dashboard (6 pages)**	Executive Overview, Spending Trends, State Performance, Drug Insights, Cost Drivers, Data Quality & Suppression	☑ Complete
7	**Executive Summary**	Business-ready summary of problem, methodology, findings, recommendations, and deliverables	☑ Complete

10. LIMITATIONS & CONSTRAINTS

Constraint	Mitigation Applied
Partial-year 2025 data	Excluded from all executive trend visuals — documented in the Data Quality dashboard page
Suppressed reimbursement records	Handled via SQL CASE logic — NULLs applied; tracked through dedicated suppression KPI views
Invalid geographic labels ('XX')	Excluded from state-level analysis — documented as unclassified jurisdictions
No therapeutic drug classification	Analysis conducted at product name level — drug category segmentation identified as a future enhancement
Gross reimbursement data only	Net spends after manufacturer rebates are not available — gross rates used throughout, noted as a limitation

11. FUTURE ENHANCEMENTS

- Predictive analytics and spend forecasting models based on historical trend patterns
- Drug therapeutic category segmentation (requires external classification data)
- Year-over-Year (YoY) growth percentage KPI for automated trend reporting
- Rebate-adjusted spend analysis — net cost after manufacturer rebates
- Beneficiary-level per-member-per-month (PMPM) cost metrics
- Automated Power BI refresh pipeline for real-time reporting
- Advanced state efficiency index — composite benchmarking metric

12. ANALYST SIGN-OFF

This Executive Summary was prepared by Bhumi Nagar as part of the Medicaid Drug Spend Intelligence & Cost Optimization project portfolio. All findings are derived from publicly available CMS Medicaid State Drug Utilization Data and are intended to support healthcare cost optimization and analytical decision-making for Medicaid managed care organizations.

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